Introduction to Machine Learning (Study Guide)

A Comprehensive Overview for LifeOS Learning companion

Topic 1: Supervised Learning concepts

Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs. It infers a function from labeled training data consisting of a set of training examples. A supervised learning algorithm analyzes the training data and produces an inferred function, which can be used for mapping new examples.

Topic 2: Unsupervised Learning concepts

Unsupervised learning is a type of algorithm that learns patterns from untagged data. The hope is that through mimicry, which is an important mode of learning in people, the machine is forced to build a compact internal representation of its world. In contrast to supervised learning, unsupervised learning algorithms are left to find structures and patterns in data without any guidance.

Topic 3: Overfitting and Underfitting

Overfitting is a concept in machine learning that occurs when a statistical model fits its training data too well. The training algorithm adjusts the weights to minimize the loss for the training set, but it captures noise and anomalies rather than the underlying data distribution. Consequently, the model performs poorly on unseen validation data.

Underfitting occurs when a machine learning model cannot capture the underlying trend of the data. It is often the result of an excessively simple model (e.g. fitting a linear model to a non-linear dataset). Underfit models show low accuracy on both training and test data.

Practice review point:

Make sure to study the trade-off between Bias and Variance, which directly relates to overfitting and underfitting.